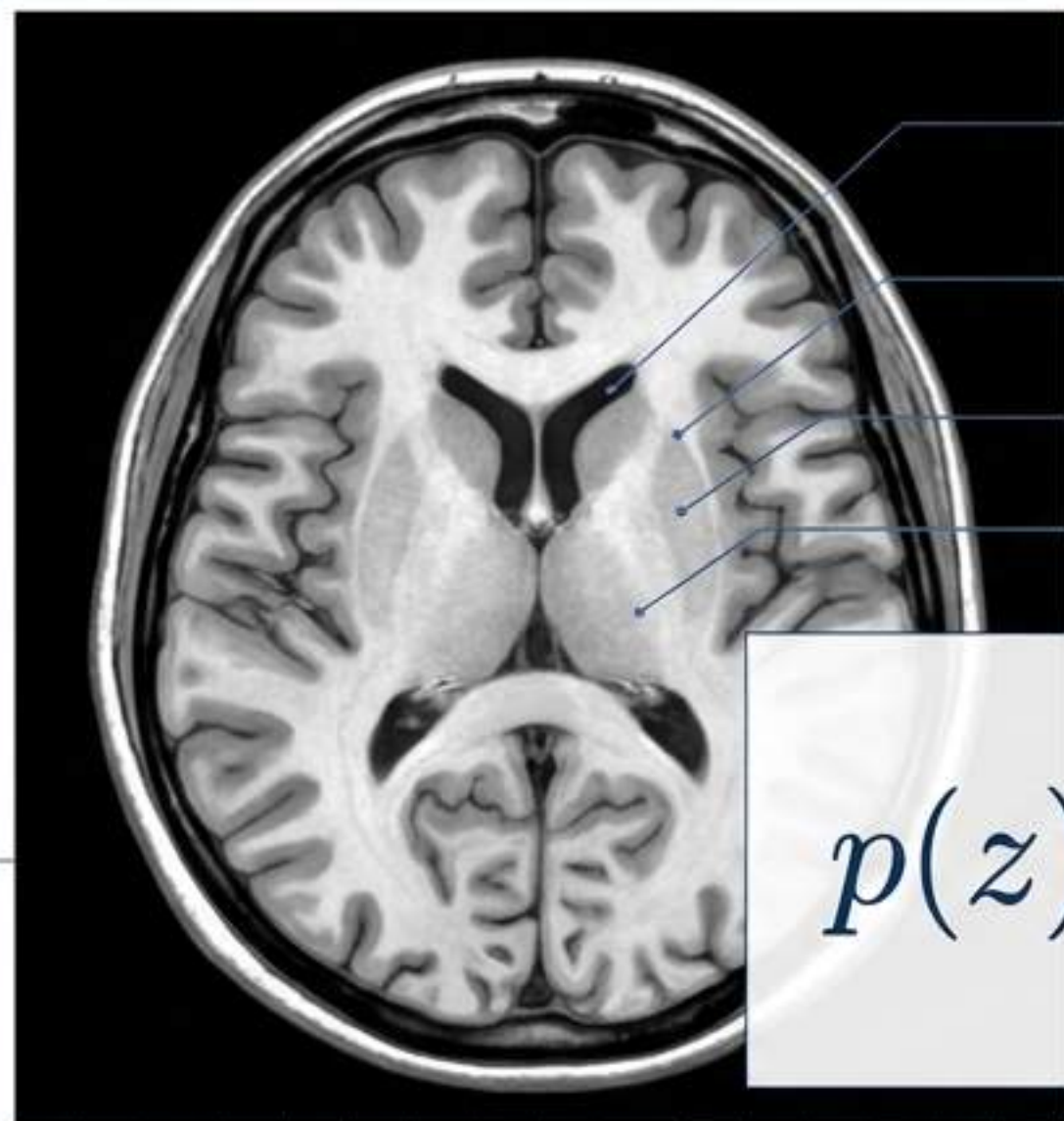




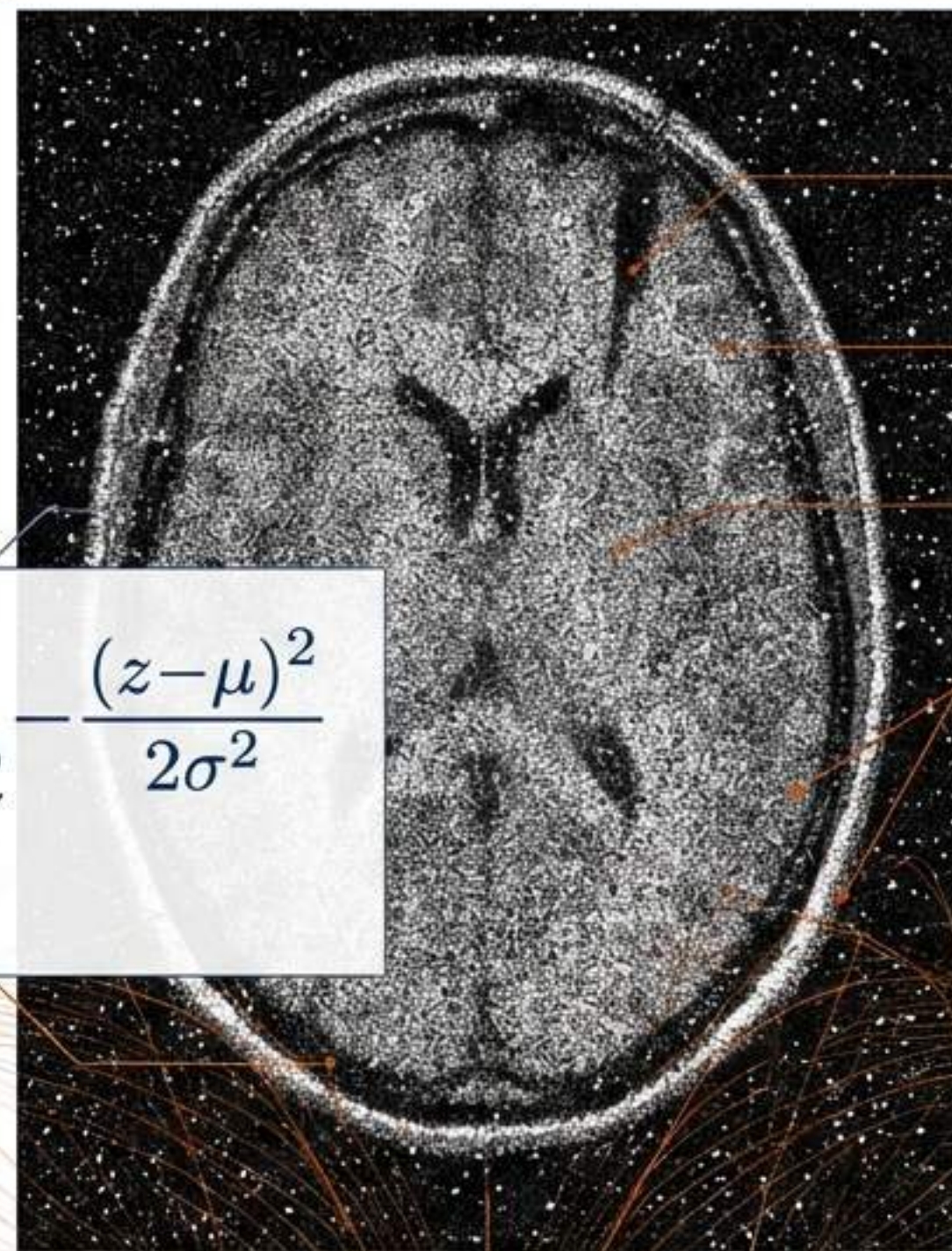
Lateral Ventricle

Caudate Nucleus

Putamen

Thalamus

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$



Signal Drop-out

Speckle Noise

Gaussian Noise Artifact

Salt-and-Pepper Noise

The Architecture of Medical Image Noise

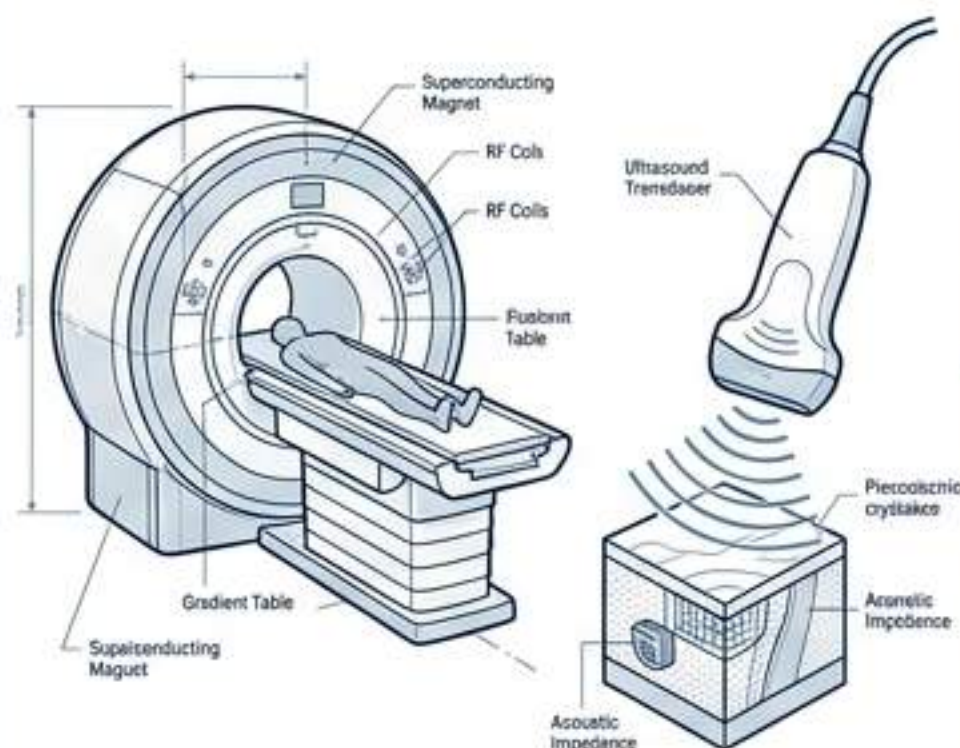
Translating Mathematical Probability Density Functions to Clinical Manifestations

Based on Gonzalez & Woods, Digital Image Processing (Chapter 5)

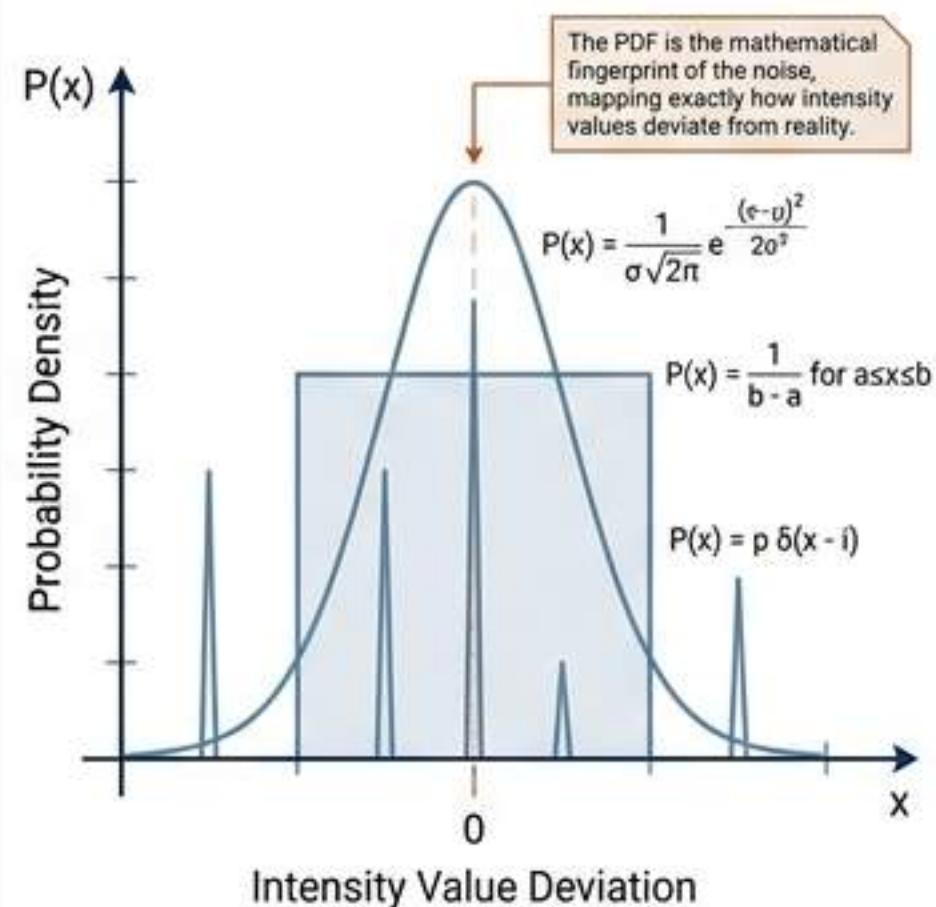
Diagnosing the Invisible Enemy

In medical imaging, noise is never truly random. It is an algorithmic artifact born from physical hardware constraints. By identifying the exact Probability Density Function (PDF) of the interference, we can design targeted algorithms to recover critical diagnostic data.

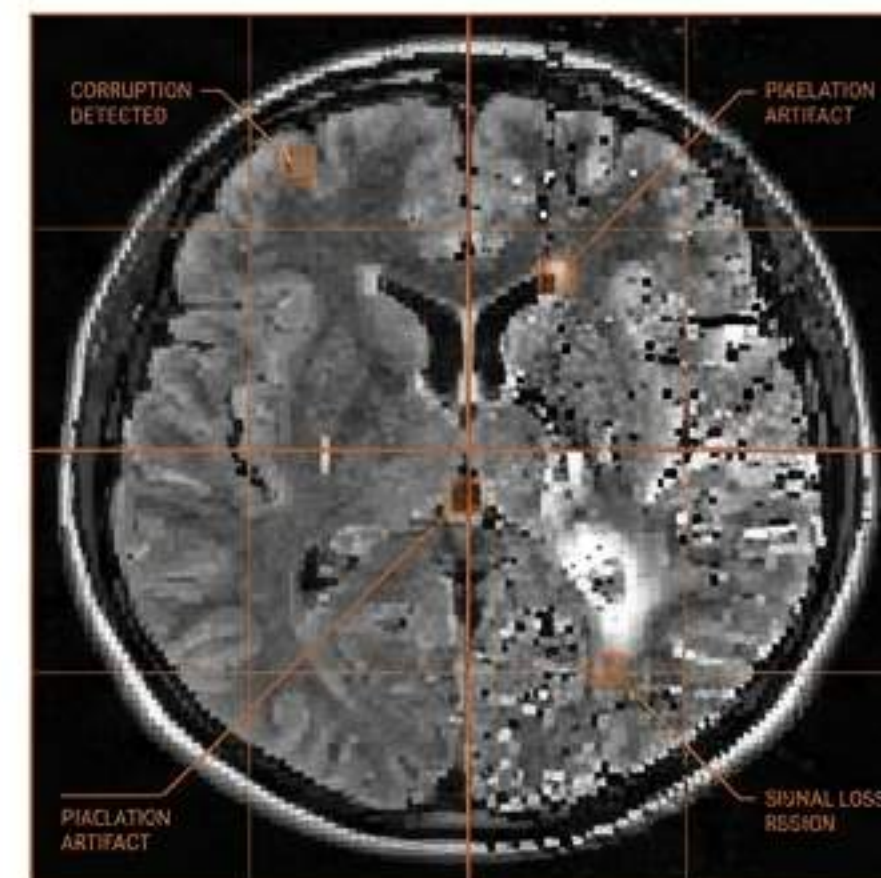
Physical Source



Mathematical Fingerprint



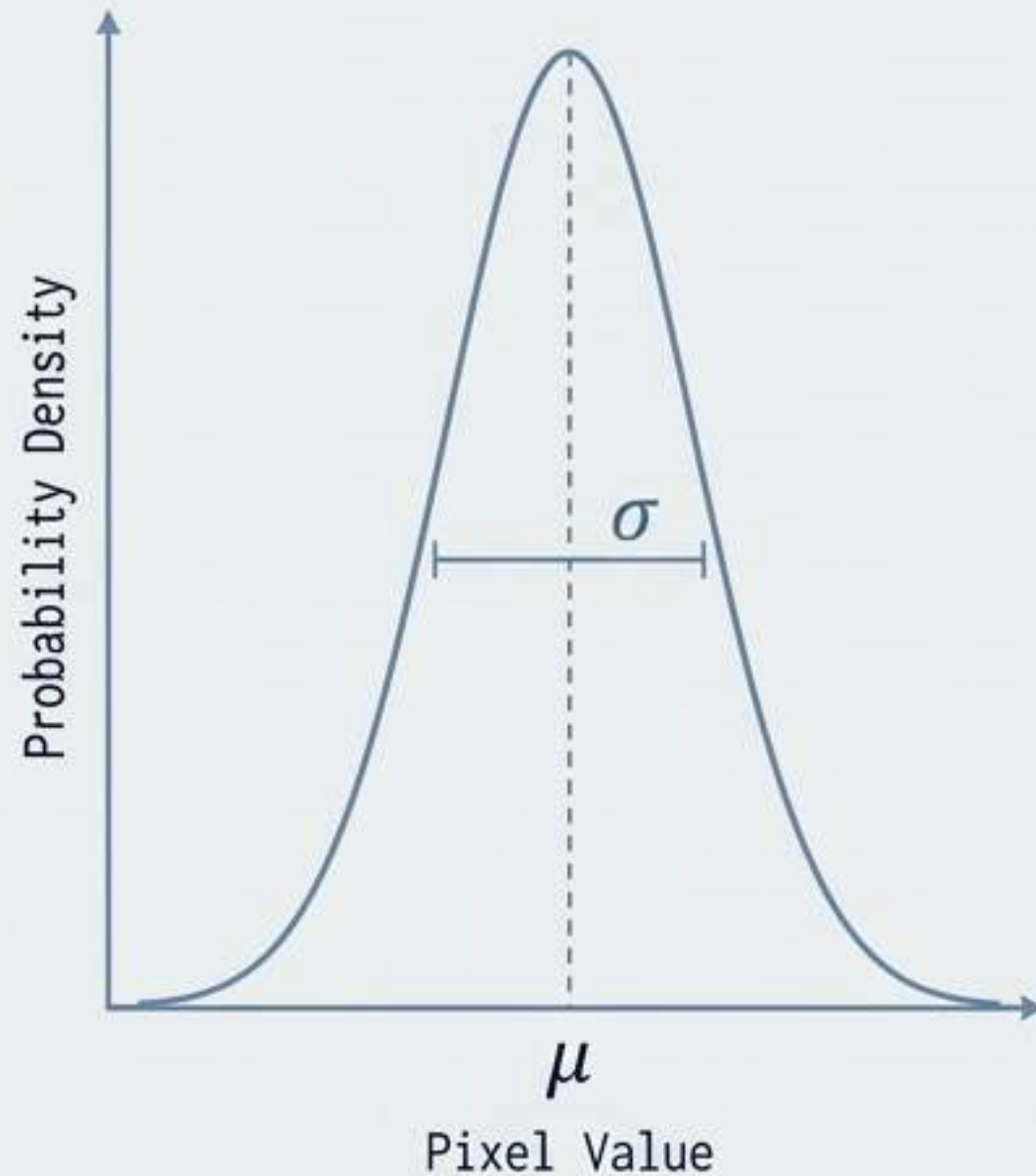
Visual Artifact



Gaussian Noise

Concept: Pixel values drift around a mean in a bell-shape. Most variations are small and continuous.

The Signature



The Blueprint

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

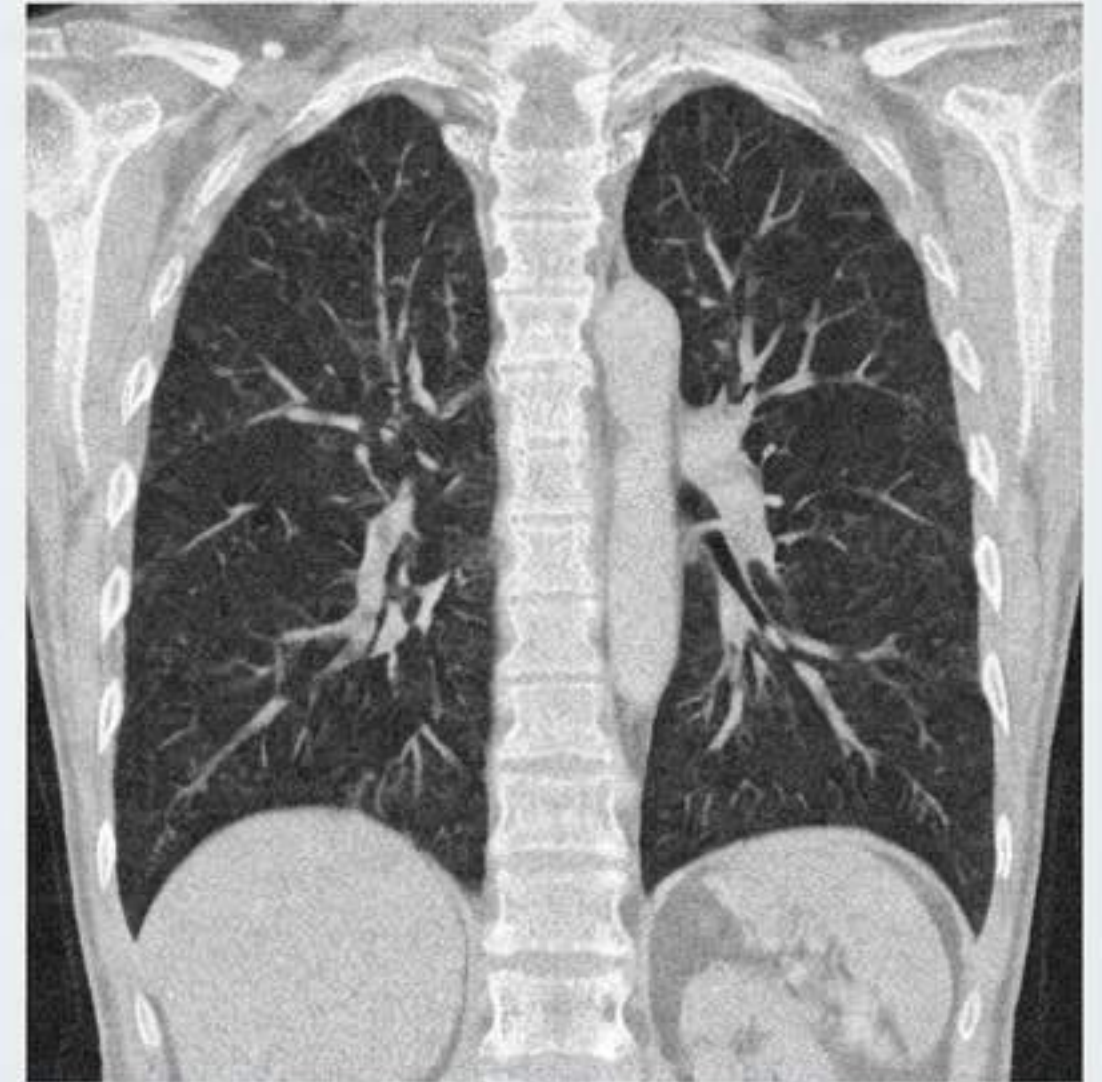
z : Intensity value
(pixel brightness)

μ : Mean of the noise
(usually assumed to be 0)

σ : Standard deviation
(larger value = stronger noise)

σ^2 : Variance

The Manifestation

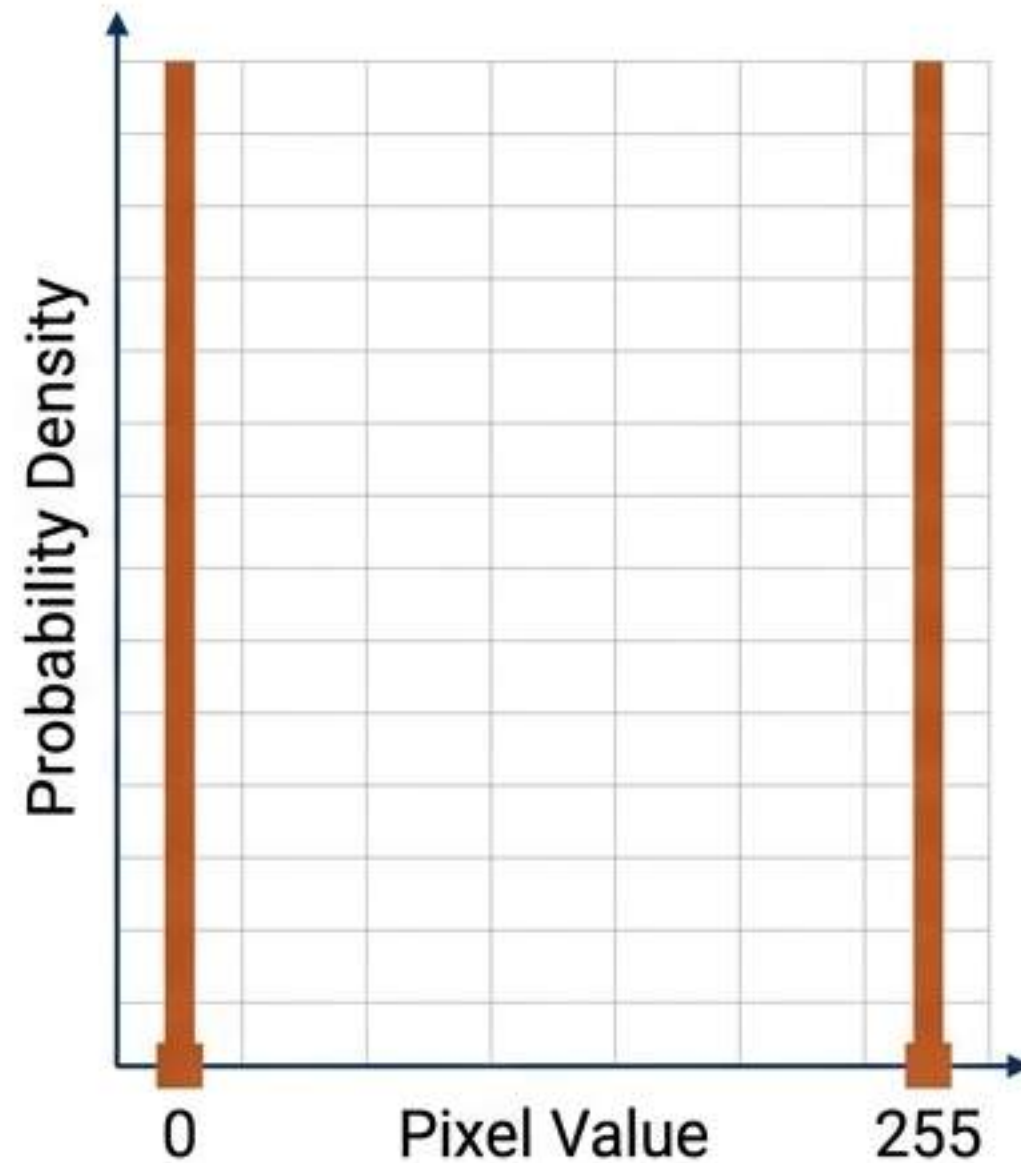


Clinical Vector: MRI & CT Scans.
Driven by ambient heat and electronic
sensor interference within the machine.

Impulse (Salt & Pepper) Noise

Concept: Pixels are not drifted; they are completely obliterated. Selected randomly, corrupted pixels are pushed to absolute black (0) or absolute white (255).

The Signature



The Blueprint

$$p(z) = \begin{cases} P_s & \text{for } z = 255 \text{ (Salt)} \\ P_p & \text{for } z = 0 \text{ (Pepper)} \end{cases}$$

P_s : Probability of a Salt strike (typically 0 to 0.1)

P_p : Probability of a Pepper strike (typically 0 to 0.1)

The Manifestation

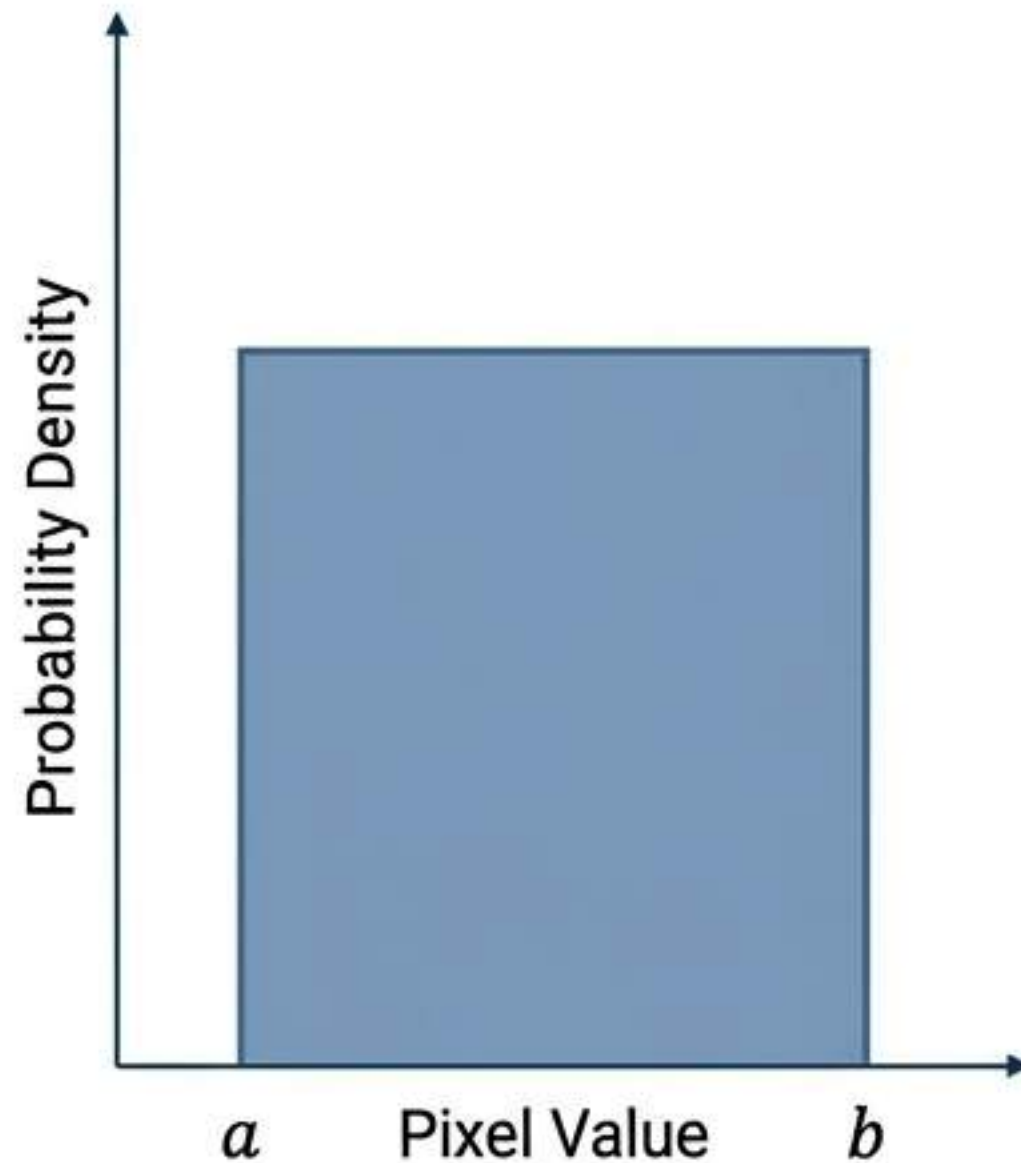


Clinical Vector: Ultrasound & PACS Systems. Caused by strong acoustic reflections bouncing back to the probe, or bit-rate errors during hospital network transmission.

Uniform Noise

Concept: Noise values are selected with exactly equal probability within a strict, defined mathematical range.
Exceedingly rare in nature.

The Signature



The Blueprint

$$p(z) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

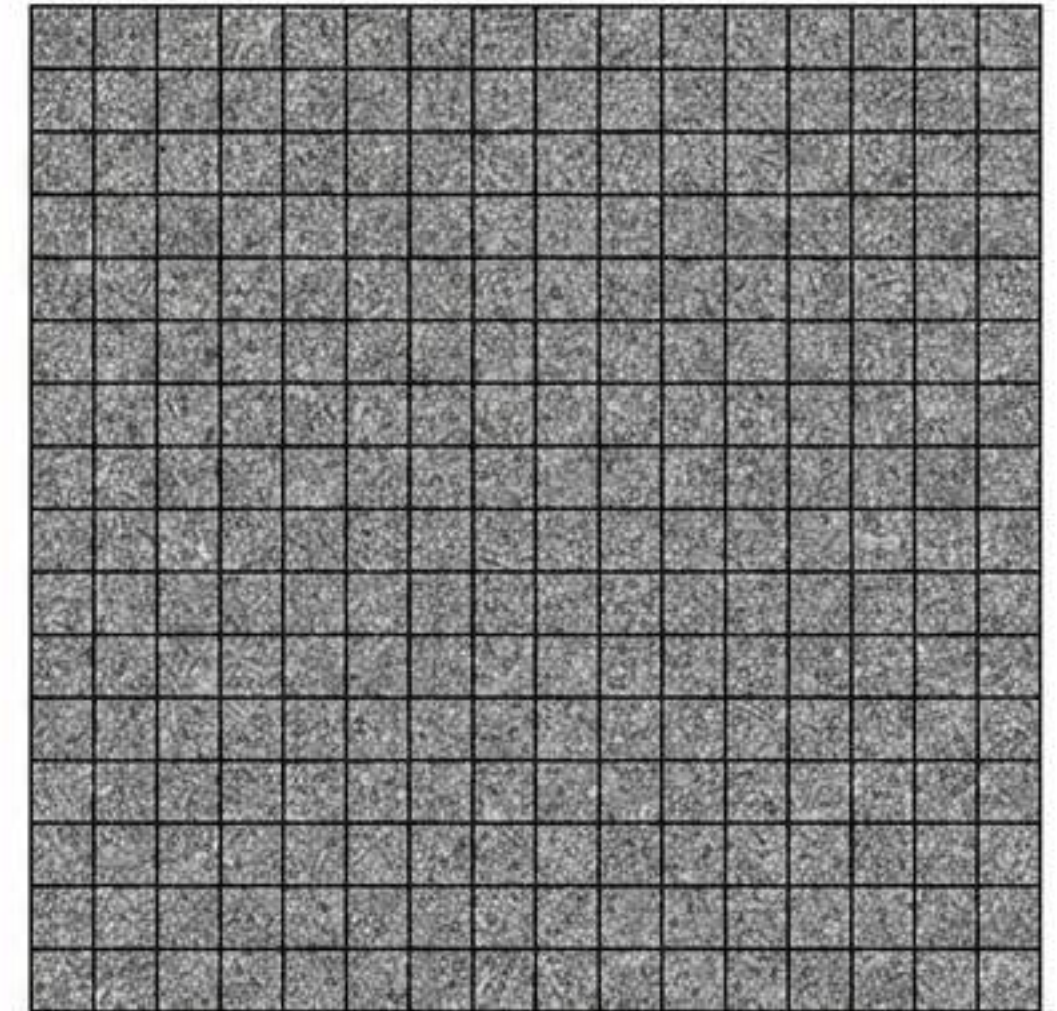
a : Lower bound of the noise range

b : Upper bound of the noise range

Mean: $(a+b)/2$

Variance: $(b-a)^2/12$

The Manifestation

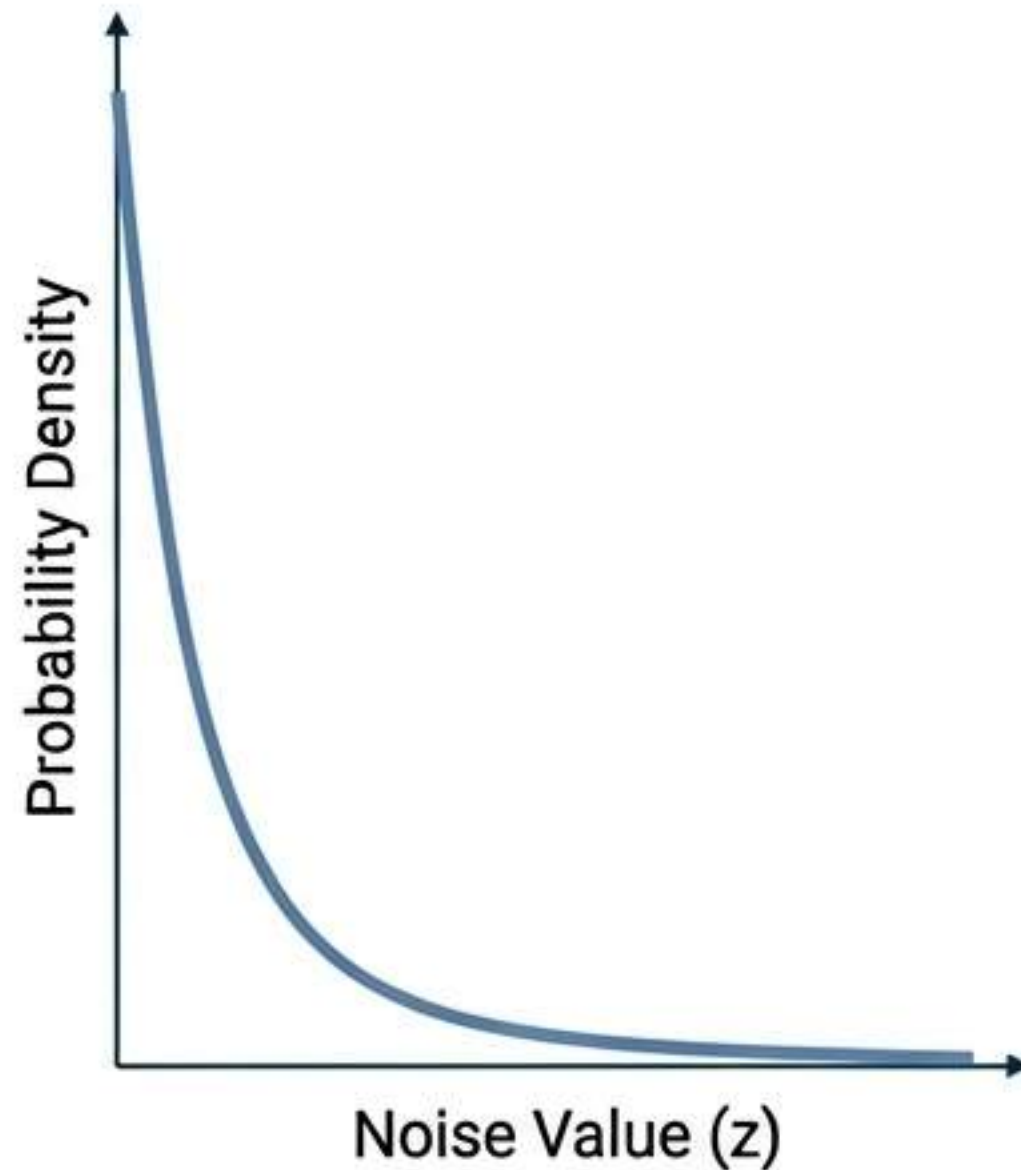


Clinical Vector: Computer Simulations. Used primarily as an artificial baseline to test and evaluate the efficacy of noise-reduction algorithms before deploying them on real patient data.

Exponential Noise

Concept: The vast majority of noise artifacts are minor, but the mathematical long tail allows for rare, exceptionally large image disruptions.

The Signature



The Blueprint

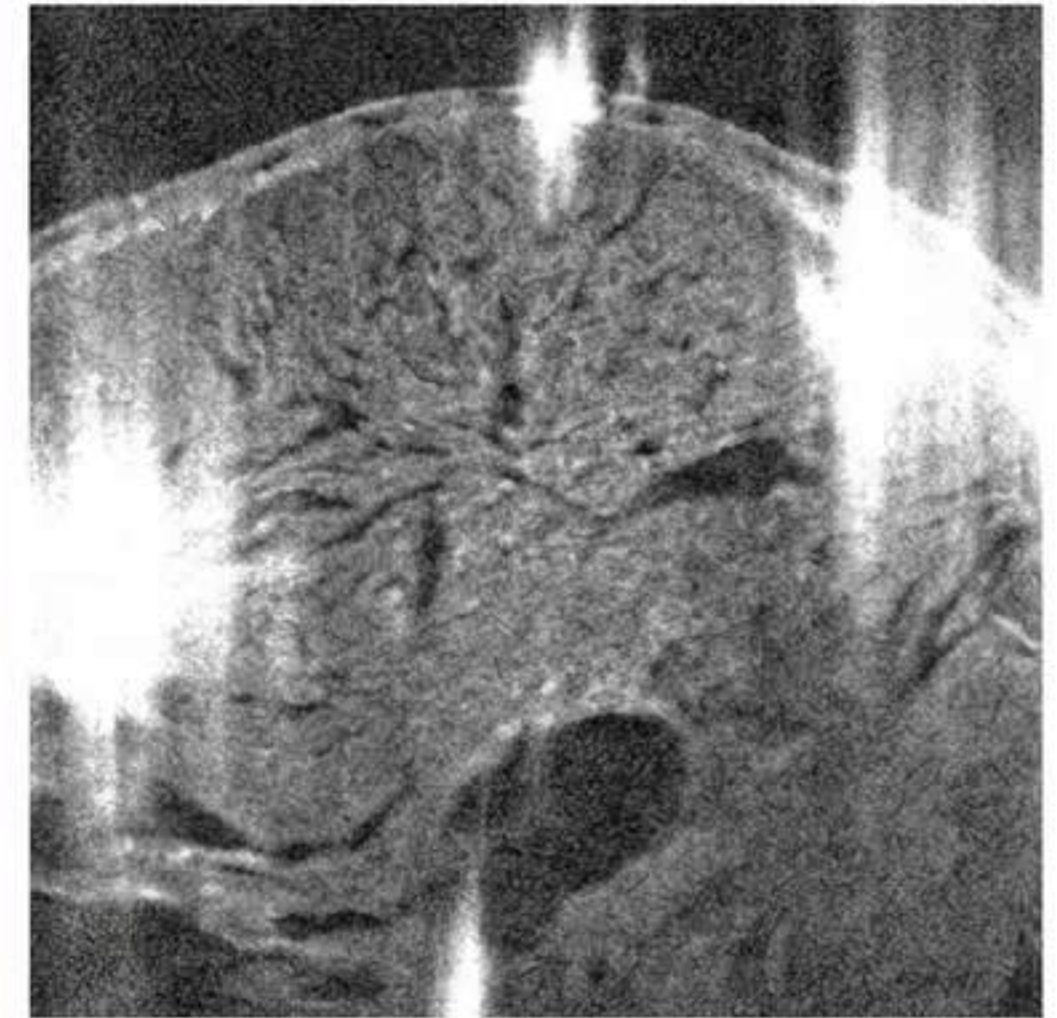
$$p(z) = \begin{cases} a \cdot e^{-a \cdot z} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

a (or λ): The rate parameter ($a > 0$)

Mean: $1/a$

Variance: $1/a^2$

The Manifestation

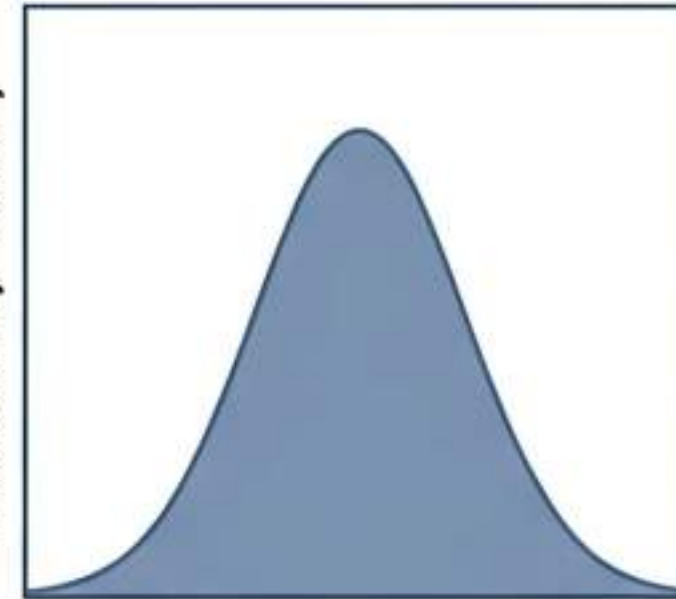


Clinical Vector: Optical Tomography & Medical Radar. Prevalent in laser-based tissue imaging and radar tracking systems for human organs.

The Mathematical Topography of Noise

Stripped of their variables, the pure Probability Density Functions reveal distinct topographical categories. The shape of the curve dictates the shape of the required algorithm.

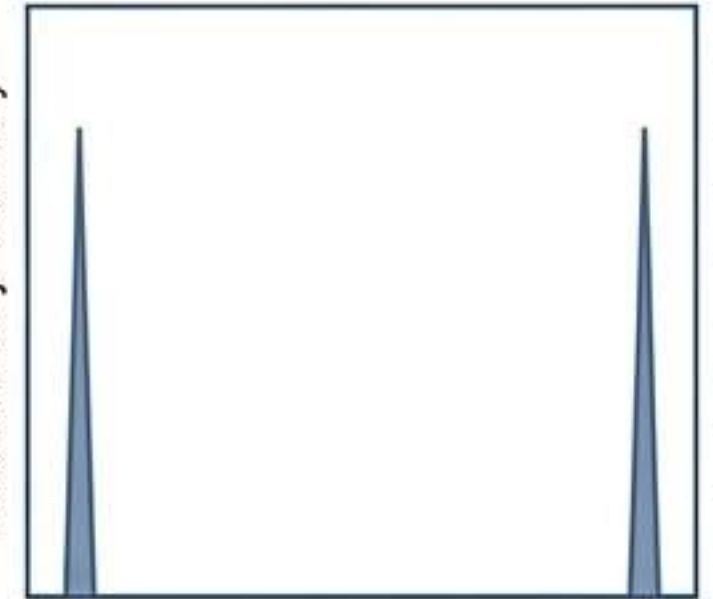
Probability Density



Pixel Value (z)

Continuous / Thermal drift

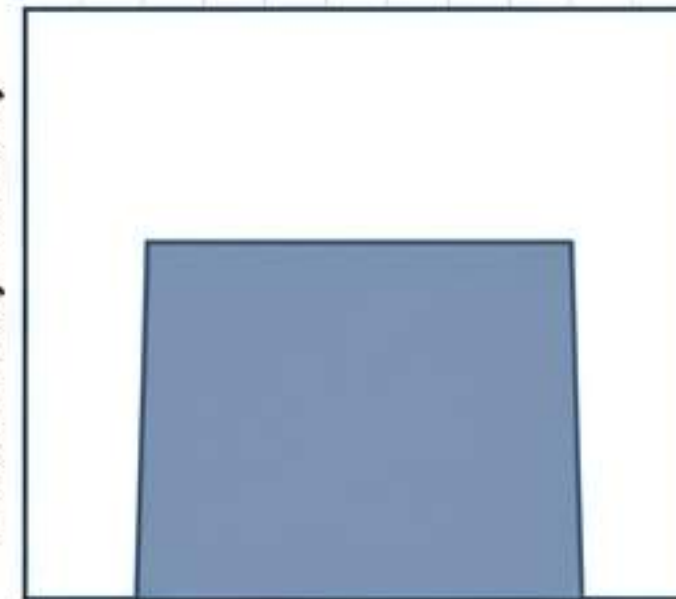
Probability Density



Pixel Value (z)

Discrete / Signal destruction

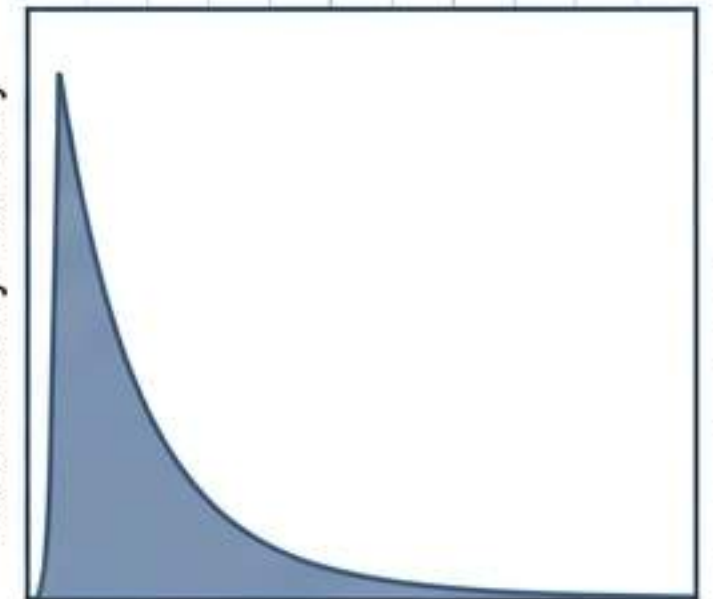
Probability Density



Pixel Value (z)

Continuous / Artificial bounds

Probability Density



Pixel Value (z)

Continuous / High-intensity flares

Diagnostic Synthesis Matrix

Noise Model	Key Parameters	Visual Signature	Primary Clinical Vector
Gaussian	μ, σ, σ^2	Smooth, bell-shaped intensity drift	MRI, CT Scans
Impulse (S&P)	P_s, P_p	Complete pixel obliteration to 0 or 255	Ultrasound, PACS Networks
Uniform	a, b	Perfect distribution within bounds	Algorithm Simulation
Exponential	$a \text{ (or } \lambda)$	Minor drift with severe, rare flares	Laser Tomography, Radar

Targeted Algorithmic Treatment

In digital image processing, noise cannot be eliminated blindly. Generic blurring destroys vital diagnostic tissue data.

By tracing the visual artifact back to its physical hardware, and matching it to its precise mathematical PDF, we engineer targeted spatial filters. We don't just erase the noise—we calculate exactly how to ignore it, preserving the life-saving data underneath.

